

**SAFETY DATA SHEET**

Creation Date 16-November-2010

Revision Date 24-December-2021

Revision Number 4

**1. Identification**

**Product Name** Barium perchlorate, anhydrous

**Cat No. :** AC221270000; AC221270050; AC221271000; AC221275000

**CAS-No** 13465-95-7  
**Synonyms** Barium Perchlorate

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company**

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

**2. Hazard(s) identification****Classification**

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                  |            |
|----------------------------------|------------|
| <b>Oxidizing solids</b>          | Category 1 |
| <b>Acute oral toxicity</b>       | Category 4 |
| <b>Acute Inhalation Toxicity</b> | Category 4 |

**Label Elements**

**Signal Word**  
Danger

**Hazard Statements**  
May cause fire or explosion; strong oxidizer  
Harmful if swallowed or if inhaled  
Harmful if inhaled

**Precautionary Statements****Prevention**

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

Keep/Store away from clothing/combustible materials

Take any precaution to avoid mixing with combustibles

Avoid breathing dust/fume/gas/mist/vapors/spray

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Use only outdoors or in a well-ventilated area

Wear protective gloves/protective clothing/eye protection/face protection

Wear fire/flammable resistant/retardant clothing

**Response**

IF INHALED: Remove person to fresh air and keep comfortable for breathing

IF ON CLOTHING: Rinse immediately contaminated clothing and skin with plenty of water before removing clothes

Call a POISON CENTER/ doctor if you feel unwell

Rinse mouth

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component          | CAS-No     | Weight % |
|--------------------|------------|----------|
| Barium perchlorate | 13465-95-7 | >95      |

### 4. First-aid measures

|                                        |                                                                                                                 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | If symptoms persist, call a physician.                                                                          |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention. |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.                       |
| <b>Inhalation</b>                      | Remove to fresh air. Get medical attention. If not breathing, give artificial respiration.                      |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.           |
| <b>Most important symptoms/effects</b> | No information available.                                                                                       |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                           |

### 5. Fire-fighting measures

|                                       |                          |
|---------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b> | No information available |
| <b>Flash Point</b>                    | No information available |

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | Not applicable           |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Oxidizing Properties</b>             | Oxidizer                 |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

**Specific Hazards Arising from the Chemical**

Oxidizer: Contact with combustible/organic material may cause fire. May ignite combustibles (wood paper, oil, clothing, etc.).

**Hazardous Combustion Products**

Thermal decomposition can lead to release of irritating gases and vapors. Hydrogen chloride gas.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
OX

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.                                                                                                              |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                                                                             |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Sweep up and shovel into suitable containers for disposal. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. Keep away from clothing and other combustible materials. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store near combustible materials. Incompatible Materials. Strong reducing agents. Combustible material. Strong oxidizing agents. Alcohols.                   |

## 8. Exposure controls / personal protection

|                            |                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b> | This product does not contain any hazardous materials with occupational exposure limit established by the region specific regulatory bodies. |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|

| Component          | Alberta                    | British Columbia           | Ontario TWAEV              | Quebec                     | ACGIH TLV                  | OSHA PEL                             | NIOSH IDLH                                               |
|--------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|--------------------------------------|----------------------------------------------------------|
| Barium perchlorate | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> | (Vacated) TWA: 0.5 mg/m <sup>3</sup> | IDLH: 50 mg/m <sup>3</sup><br>TWA: 0.5 mg/m <sup>3</sup> |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Ventilation systems. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate

ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

**Eye Protection**  
**Hand Protection**

Goggles  
Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                               |                                                   |
|-----------------------------------------------|---------------------------------------------------|
| <b>Physical State</b>                         | Powder Solid                                      |
| <b>Appearance</b>                             | White                                             |
| <b>Odor</b>                                   | Odorless                                          |
| <b>Odor Threshold</b>                         | No information available                          |
| <b>pH</b>                                     | No information available 5.6 100g/l aq.sol.(20°C) |
| <b>Melting Point/Range</b>                    | 505 °C / 941 °F                                   |
| <b>Boiling Point/Range</b>                    | No information available                          |
| <b>Flash Point</b>                            | No information available                          |
| <b>Evaporation Rate</b>                       | Not applicable                                    |
| <b>Flammability (solid,gas)</b>               | No information available                          |
| <b>Flammability or explosive limits</b>       |                                                   |
| Upper                                         | No data available                                 |
| Lower                                         | No data available                                 |
| <b>Vapor Pressure</b>                         | No information available                          |
| <b>Vapor Density</b>                          | Not applicable                                    |
| <b>Specific Gravity</b>                       | 3.200                                             |
| <b>Solubility</b>                             | No information available                          |
| <b>Partition coefficient; n-octanol/water</b> | No data available                                 |
| <b>Autoignition Temperature</b>               | Not applicable                                    |
| <b>Decomposition Temperature</b>              | > 700°C                                           |
| <b>Viscosity</b>                              | Not applicable                                    |
| <b>Molecular Formula</b>                      | Ba Cl <sub>2</sub> O <sub>8</sub>                 |

Molecular Weight

336.23

## 10. Stability and reactivity

|                                         |                                                                                                     |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | Yes                                                                                                 |
| <b>Stability</b>                        | Stable under normal conditions. Oxidizer: Contact with combustible/organic material may cause fire. |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Avoid dust formation. Combustible material.                     |
| <b>Incompatible Materials</b>           | Strong reducing agents, Combustible material, Strong oxidizing agents, Alcohols                     |
| <b>Hazardous Decomposition Products</b> | Thermal decomposition can lead to release of irritating gases and vapors, Hydrogen chloride gas     |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                            |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                       |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | No information available                                                                 |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component          | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------|------------|------------|------------|------------|------------|------------|
| Barium perchlorate | 13465-95-7 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN1447  
Hazard Class 5.1  
Subsidiary Hazard Class 6.1  
Packing Group II

#### TDG

UN-No UN1447  
Hazard Class 5.1  
Subsidiary Hazard Class 6.1  
Packing Group II

#### IATA

UN-No UN1447  
Proper Shipping Name BARIUM PERCHLORATE, SOLID  
Hazard Class 5.1  
Subsidiary Hazard Class 6.1  
Packing Group II

#### IMDG/IMO

UN-No UN1447  
Proper Shipping Name BARIUM PERCHLORATE, SOLID  
Hazard Class 5.1  
Subsidiary Hazard Class 6.1  
Packing Group II

### 15. Regulatory information

#### International Inventories

| Component          | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|--------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Barium perchlorate | 13465-95-7 | X   | -    | X    | ACTIVE                                        | 236-710-4 | -      | -   |

| Component          | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Barium perchlorate | 13465-95-7 | X     | KE-02082 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

| Component          | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Barium perchlorate | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component          | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Barium perchlorate | 13465-95-7 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component          | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Barium perchlorate | 13465-95-7 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

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Thermo Fisher Scientific  
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**Revision Summary** This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**